Chapters 1 and 2: Pitch and Intervals

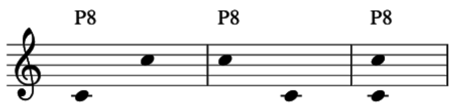
All pitches within a one-octave span.

Dyads are pairs of pitches played either in sequence (melodic) or simultaneously (harmonic). They are named after the distance between the two pitches. This distance is called an interval. Intervals are measured in half-steps, the smallest possible distance between two pitches.

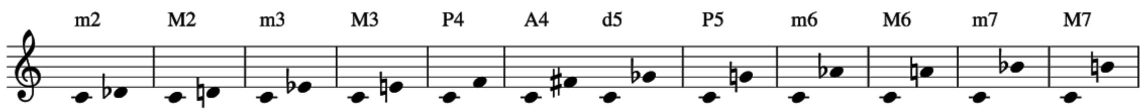
Number of half-steps between pitches	Full interval name	Abbreviation	Scale Degree Function
0	Perfect Unison	P1	1
1	Minor 2nd	m2	b2 / b9
2	Major 2nd	M2	2 / 9
3	Minor 3rd Augmented 2nd	m3 A2	b3 #2 / #9
4	Major 3rd	M3	3
5	Perfect 4th	P4	4 / 11
6	Tritone Augmented 4th Diminished 5th	TT A4 d5	 #4 / #11 b5
7	Perfect 5th	P5	5
8	Minor 6th Augmented 5th	m6 A5	b6 / b13 #5
9	Major 6th Diminished 7th	M6 d7	6 / 13 bb7
10	Minor 7th	m7	b7
11	Major 7th	M7	7
12	Perfect Octave	P8	1

Unisons occur when the same pitch is played twice.

Octaves occur when two pitches with the same name are played one octave apart.



Melodic Intervals: Ascending



Melodic Intervals: Descending

Harmonic Intervals

Interval Nomenclature

There are two components to an interval's name: a quality followed by a number.

The number indicates how many letter names are spanned between the two pitches, inclusive.

In this example, A ascends to G. Seven letter names are spanned: A-B-C-D-E-F-G. Therefore, this interval is some kind of 7th.

In this example, A descends to G. Only two letter names are spanned: A-G. Therefore, this interval is some kind of 2nd.

Qualities include Perfect, Minor, Major, Diminished, and Augmented.

Unisons, 4ths, 5ths, and Octaves can be Diminished, Perfect, or Augmented.
2nds, 3rds, 6ths, and 7ths can be Diminished, Minor, Major, or Augmented.

The table below summarizes which intervals are Perfect and which are Minor/Major, their abbreviations, and how the quality is transformed when the interval is increased or decreased by a half-step.

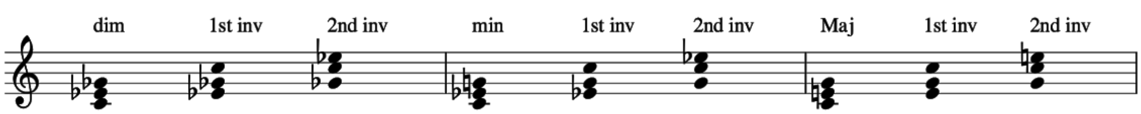
Interval Quality	Abbreviation (case-sensitive)	+1 Half-Step	-1 Half-Step
Perfect (1, 4, 5, 8)	P	Augmented	Diminished
Minor (2, 3, 6, 7)	m	Major	Diminished
Major (2, 3, 6, 7)	M	Augmented	Minor
Diminished	d	Perfect or Minor	
Augmented	A		Perfect or Major

Chapter 3a: Basic Triads

Triads contain three different pitches. Basic Triads are built by stacking pairs of third intervals. The lowest pitch in one of these stacks is called the Root. The pitch a third above the Root is called the Third, and the pitch a fifth above the Root is called the Fifth.

Triad Type	Interval: Root and Third	Interval: Third and Fifth	Interval: Root and Fifth	Formula
Diminished, d, °	m3	m3	d5	1 b3 b5
Minor, m	m3	M3	P5	1 b3 5
Major, M	M3	m3	P5	1 3 5
Augmented, A, +	M3	M3	A5	1 3 #5

Inversions are generated by changing which pitch appears in the low voice. A triad is in first inversion when the Third is in the low voice. A triad is in second inversion when the Fifth is in the low voice. A consequence of inversion is that the various intervals between the three voices (Low, Middle, and High) change.



Triad Type	Inversion	Interval: Low and Middle	Interval: Middle and High	Interval: Low and High	Formula
Diminished	1st	m3	A4	M6	b3 b5 1
Diminished	2nd	A4	m3	M6	b5 1 b3
Minor	1st	M3	P4	M6	b3 5 1
Minor	2nd	P4	m3	m6	5 1 b3
Major	1st	m3	P4	m6	3 5 1
Major	2nd	P4	M3	M6	5 1 3

Chapter 3b: Advanced Triads

Advanced Triads are built by stacking pairs of fourth intervals. These are called Quartal triads.

Triad Type	Interval: Low and Middle	Interval: Middle and High	Interval: Low and High
Q	P4	P4	m7
Q+	P4	A4	M7
+4Q	A4	P4	M7

Each inversion is treated as its own unique triad, and its lowest note is reinterpreted as a new Root.

Triad Type	Inversion	Name	Formula	Interval: Low and Middle	Interval: Middle and High	Interval: Low and High
Q	1st	sus4	1 4 5	P4	M2	P5
Q	2nd	sus2	1 2 5	M2	P4	P5
Q+	1st	Lydian	1 #4 5	A4	m2	P5
Q+	2nd	Locrian	1 b2 b5	m2	P4	d5
+4Q	1st	sus4b5	1 4 b5	P4	m2	d5
+4Q	2nd	Phrygian	1 b2 5	m2	A4	P5

Chapter 4: Sevenths

Seventh chords are formed when a 7th is added above a Triad’s Root.

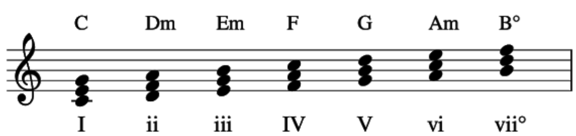
Base Triad	Added 7th	Name of 7th Chord	Formula
Major	M7	Major 7	1 3 5 7
Major	m7	7 Dominant 7	1 3 5 b7
Minor	M7	Minor Major 7	1 b3 5 7
Minor	m7	Minor 7	1 b3 5 b7
Diminished	M7	Diminished Major 7	1 b3 b5 7
Diminished	m7	Half-diminished 7 Minor 7 b5	1 b3 b5 b7
Diminished	d7	Diminished 7	1 b3 b5 bb7
Augmented	M7	Augmented Major 7	1 3 #5 7
sus4	M7	Major 7 sus4	1 4 5 7
sus4	m7	7sus4 Dominant 7 sus4	1 4 5 b7
sus2	M7	Major 7 sus2	1 2 5 7
Lydian	M7	Lydian Major 7	1 #4 5 7
Lydian	m7	Lydian Dominant 7	1 #4 5 b7
Phrygian	M7	Phrygian Major 7	1 b2 5 7
Phrygian	m7	Phrygian Dominant 7	1 b2 5 b7

Chapter 5: Tonal Progressions

The Major scale is a set of seven pitches built on any given Root.

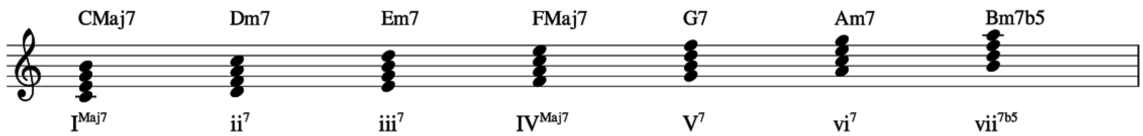
Scale Degree Number	Scale Degree Name	Interval Above Root
1	Tonic	P1
2	Supertonic	M2
3	Mediant	M3
4	Subdominant	P4
5	Dominant	P5
6	Submediant	M6
7	Leading tone	M7

A triad can be built in diatonic thirds on each scale degree. “Diatonic” means that only pitches from the scale are used.



Scale Degree	Triad Quality	Roman Numeral	Function
1	Major	I	Tonic
2	Minor	ii	Pre-Dominant
3	Minor	iii	Tonic
4	Major	IV	Pre-Dominant
5	Major	V	Dominant
6	Minor	vi	Tonic
7	Diminished	vii°	Dominant

A seventh chord can be built in diatonic thirds on each scale degree.

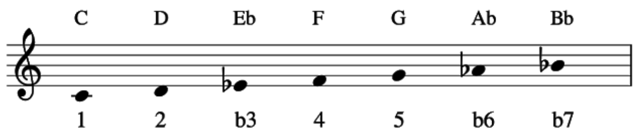


Scale Degree	Seventh Quality	Roman Numeral	Function
1	Major 7	I^{Maj7}	Tonic
2	Minor 7	ii^{7}	Pre-Dominant
3	Minor 7	iii^{7}	Tonic
4	Major 7	IV^{Maj7}	Pre-Dominant
5	Dominant 7	V^{7}	Dominant
6	Minor 7	vi^{7}	Tonic
7	Half-diminished Minor 7 b5	$vii^{ø7}$	

Roman Numeral Nomenclature

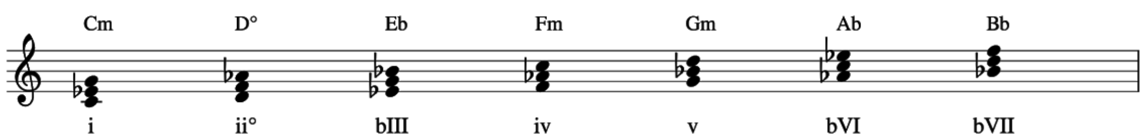
Chord Type	Case	Additional Symbol
Diminished	Lower-Case	°
Minor	Lower-Case	
Major	Upper-Case	
Augmented	Upper-Case	+

The Minor scale is a set of seven pitches built on any given Root.
Note that compared to the Major scale, the third, sixth, and seventh scale degrees are flattened.



Scale Degree Number	Scale Degree Name	Interval Above Root
1	Tonic	P1
2	Supertonic	M2
b3	Mediant	m3
4	Subdominant	P4
5	Dominant	P5
b6	Submediant	m6
b7	Subtonic	m7

A triad can be built in diatonic thirds on each scale degree.



Scale Degree	Triad Quality	Roman Numeral
1	Minor	i
2	Diminished	ii°
b3	Major	bIII
4	Minor	iv
5	Minor	v
b6	Major	bVI
b7	Major	bVII

Tonal Progressions Minor Key

The progressions in this module combine chords in Root Position with chords in 1st Inversion. This means that a given bass note can potentially support several chords.

Scale Degree in the Bass	Root Position Chord	1st Inversion Chord
1	i	bVI
2	ii°	bVII
b3	bIII	i
4	iv	ii°
5	v	bIII
5	V	
b6	bVI	iv
b7	bVII	v

Chapter 6: Secondary Chords

In a Major key, secondary dominant chords are the V7 and vii°7 chords of ii, iii, IV, V, and vi.

x	V7/x	vii°7/x
Dm (ii)	A7	C#°7
Em (iii)	B7	D#°7
F (IV)	C7	E°7
G (V)	D7	F#°7
Am (vi)	E7	G#°7

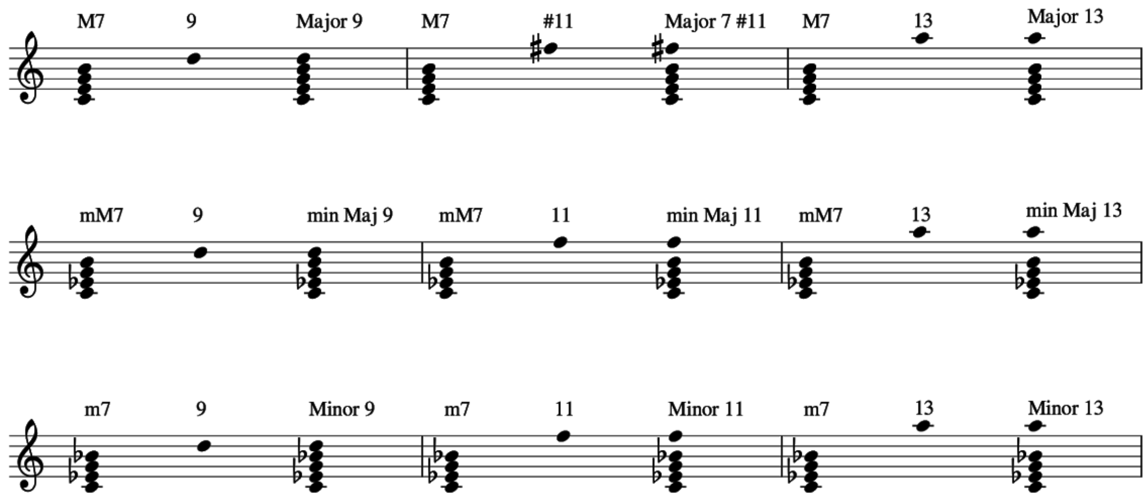
Chapter 7a: Added-Note Chords

Added-Note Chords are formed when a 2nd, 4th, or 6th is added above a Triad's Root.

Base Triad	Added 2, 4, or 6	Name of New Chord	Formula
Major	m2	Phrygian Major	1 b2 3 5
Major	M2	Major add 2	1 2 3 5
Major	P4	Major add 4	1 3 4 5
Major	A4	Lydian Major	1 3 #4 5
Major	m6	Major add b6	1 3 5 b6
Major	M6	Major 6	1 3 5 6
Minor	m2	Phrygian Minor	1 b2 b3 5
Minor	M2	Minor add 2	1 2 b3 5
Minor	P4	Minor add 4	1 b3 4 5
Minor	A4	Lydian Minor	1 b3 #4 5
Minor	m6	Minor add b6	1 b3 5 b6
Minor	M6	Minor 6	1 b3 5 6

Chapter 7b: Extended Chords

Extended Chords are formed when a 9th, 11th, or 13th is added above a Seventh chord's Root.
Note: 9ths, 11ths, and 13ths are equivalent to 2nds, 4ths, and 6ths.



Base Seventh Chord	Added 9, 11, or 13	Name of Chord	Formula
Major 7	M9	Major 9	1 3 5 7 9
Major 7	A11	Major 7 #11	1 3 5 7 #11
Major 7	M13	Major 13	1 3 5 7 13
Minor 7	M9	Minor 9	1 b3 5 b7 9
Minor 7	P11	Minor 11	1 b3 5 b7 11
Minor 7	M13	Minor 13	1 b3 5 b7 13
Minor Major 7	M9	Minor Major 9	1 b3 5 7 9
Minor Major 7	P11	Minor Major 11	1 b3 5 7 11
Minor Major 7	M13	Minor Major 13	1 b3 5 7 13

Chapter 8: Rhythm

x note-heads in the sheet music, and shaded cells in the graphs denote the click.

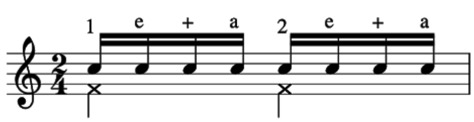
Quarters

1	2	3	4
5	6	7	8

Eighths

1	+	2	+
3	+	4	+

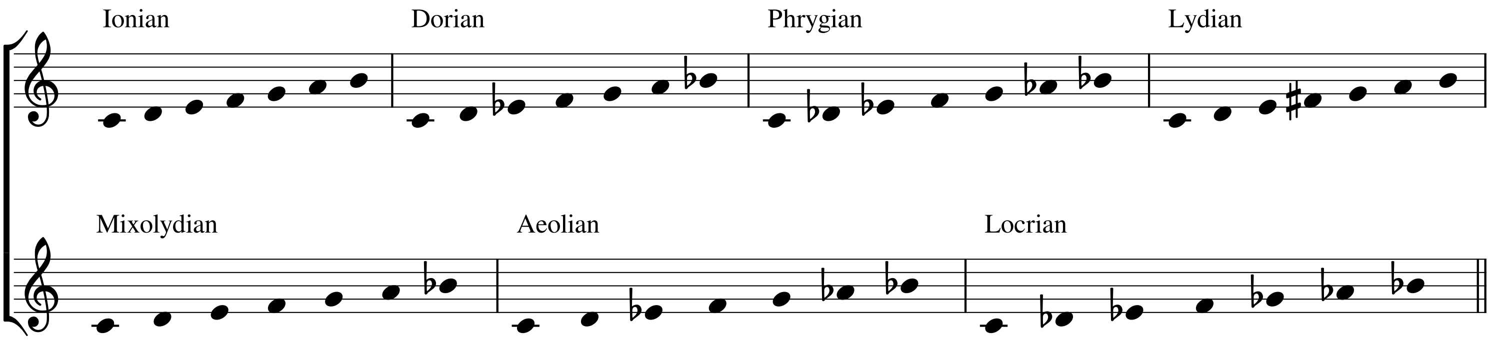
Sixteenths



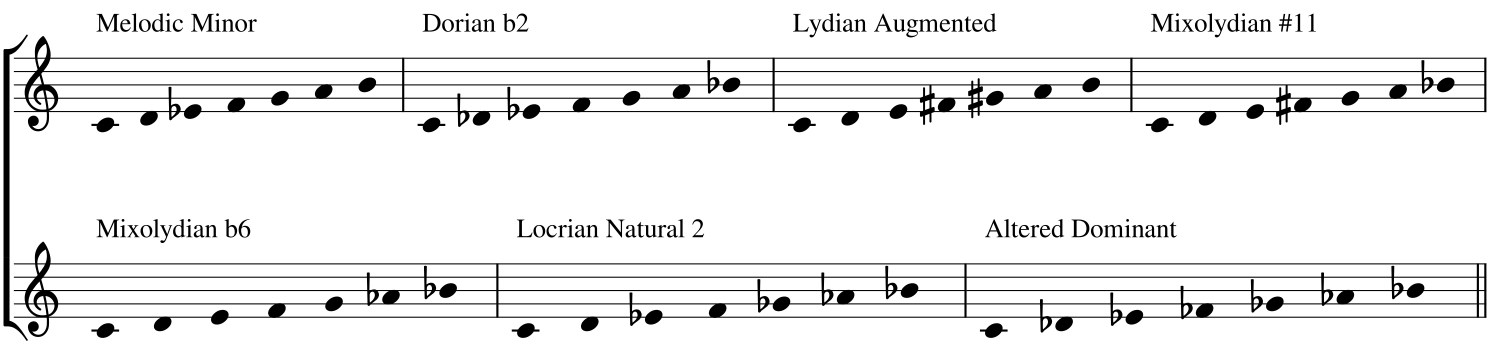
1	e	+	a
2	e	+	a

Chapter 9: Scales

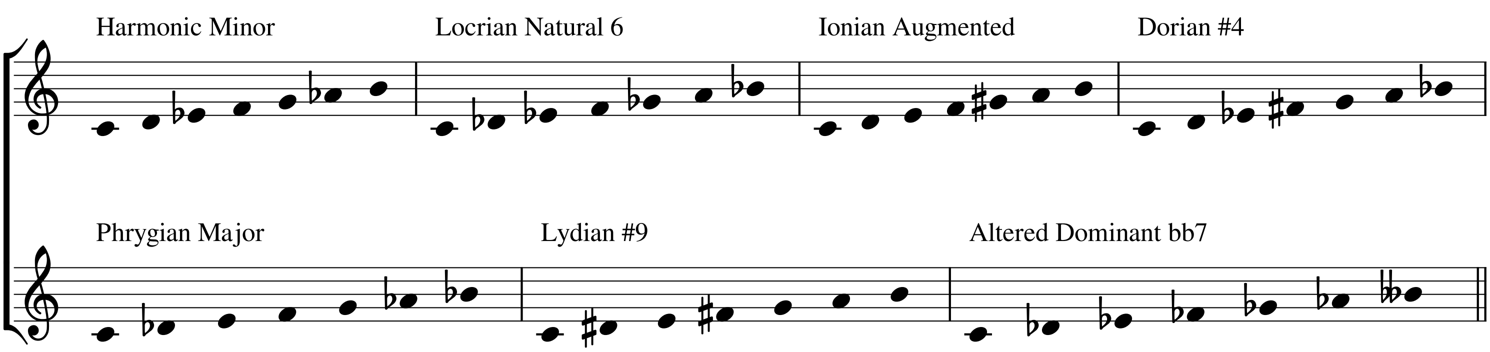
Major



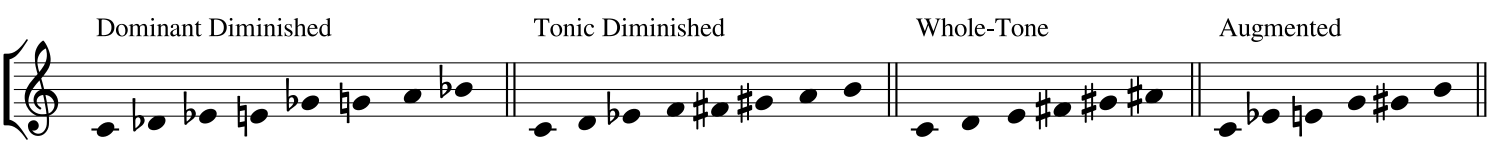
Melodic Minor



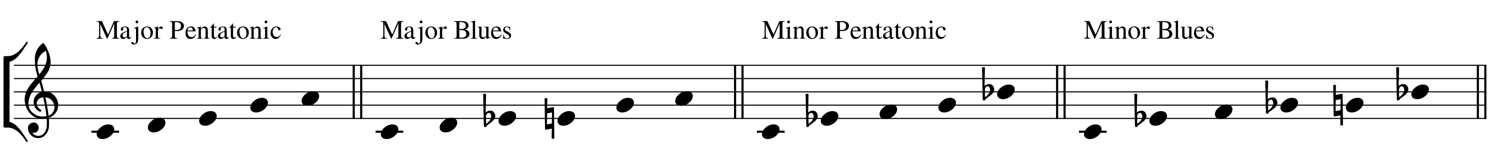
Harmonic Minor



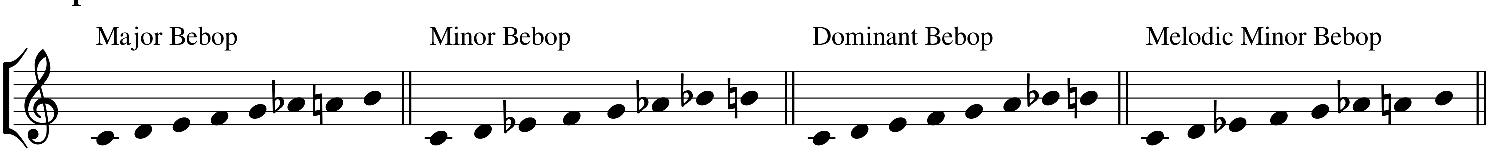
Symmetrical



Pentatonic/Blues



Bebop



Modes of the Major Scale

Ionian	1		2		3	4		5		6		7
Dorian	1		2	b3		4		5		6	b7	
Phrygian	1	b2		b3		4		5	b6		b7	
Lydian	1		2		3		#4	5		6		7
Mixolydian	1		2		3	4		5		6	b7	
Aeolian	1		2	b3		4		5	b6		b7	
Locrian	1	b2		b3		4	b5		b6		b7	

Modes of the Melodic Minor Scale

Melodic Minor	1		2	b3		4		5		6		7
Dorian b2	1	b2		b3		4		5		6	b7	
Lydian Augmented	1		2		3		#4		#5	6		7
Mixolydian #11	1		2		3		#4	5		6	b7	
Mixolydian b6	1		2		3	4		5	b6		b7	
Locrian Natural 2	1		2	b3		4	b5		b6		b7	
Altered Dominant	1	b2		b3 #2	b4 3		b5		b6 #5		b7	

Modes of the Harmonic Minor Scale

Harmonic Minor	1		2	b3		4		5	b6			7
Locrian Natural 6	1	b2		b3		4	b5			6	b7	
Ionian Augmented	1		2		3	4			#5	6		7
Dorian #4	1		2	b3			#4	5		6	b7	
Phrygian Major	1	b2			3	4		5	b6		b7	
Lydian #9	1			#2	3		#4	5		6		7
Altered Dominant bb7	1	b2		b3 #2	b4 3		b5 #4		b6 #5	bb7 6		

Symmetrical Scales

Dominant Diminished	1	b2		#2	3		#4	5		6	b7	
Tonic Diminished	1		2	b3		4	b5		#5	6		7
Whole Tone	1		2		3		#4		#5		b7	
Augmented	1			b3	3			5	#5			7

Pentatonic and Blues Scales

Major Pentatonic	1		2		3			5		6		
Major Blues	1		2	#2 b3	3			5		6		
Minor Pentatonic	1			b3		4		5			b7	
Minor Blues	1			b3		4	#4 b5	5			b7	

Bebop Scales

Major Bebop	1		2		3	4		5	#5 b6	6		7
Minor Bebop	1		2	b3		4		5	b6		b7	7
Dominant Bebop	1		2		3	4		5		6	b7	7
Melodic Minor Bebop	1		2	b3		4		5	#5 b6	6		7

Chapter 10: Modal Voicings

Modes are compared side-by-side. Shaded cells indicate the differences between modes.

Ionian	1		2		3	4		5		6		7
Lydian	1		2		3		#4	5		6		7

Dorian	1		2	b3		4		5		6	b7	
Aeolian	1		2	b3		4		5	b6		b7	

Phrygian	1	b2		b3		4		5	b6		b7	
Locrian	1	b2		b3		4	b5		b6		b7	

Ionian	1		2		3	4		5		6		7
Mixolydian	1		2		3	4		5		6	b7	

Melodic Minor	1		2	b3		4		5		6		7
Locrian Natural 2	1		2	b3		4	b5		b6		b7	

Dorian b2	1	b2		b3		4		5		6	b7	
Altered Dominant	1	b2		b3 #2	b4 3		b5		b6 #5		b7	

Lydian	1		2		3		#4	5		6		7
Lydian Augmented	1		2		3		#4		#5	6		7

Mixolydian #11	1		2		3		#4	5		6	b7	
Mixolydian b6	1		2		3	4		5	b6		b7	

Harmonic Minor	1		2	b3		4		5	b6			7
Phrygian Major	1	b2			3	4		5	b6		b7	

Altered Dominant	1	b2		b3 #2	b4 3		b5		b6 #5		b7	
Altered Dominant bb7	1	b2		b3 #2	b4 3		b5		b6 #5	bb7 6		

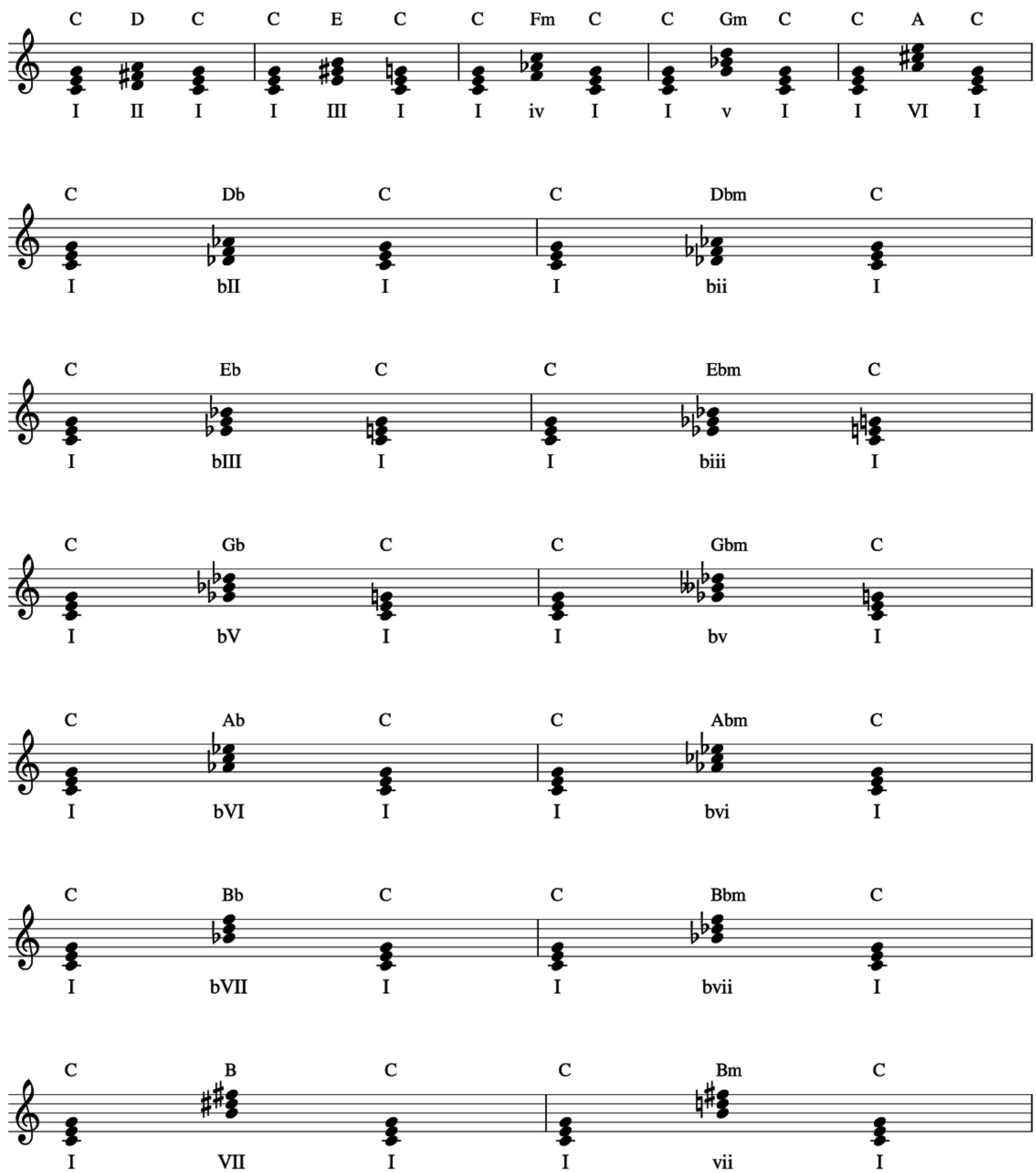
Locrian Natural 6	1	b2		b3		4	b5			6	b7	
Dorian #4	1		2	b3			#4	5		6	b7	

Ionian Augmented	1		2		3	4			#5	6		7
Lydian #9	1			#2	3		#4	5		6		7

Chapter 12: Modal Harmony

Modal Triads include all non-diatonic Major and Minor triads.

Major Tonic



Modal Triads in C Major

Roman Numeral	Triad
bII	Db
bii	Dbm
II	D
bIII	Eb
biii	Ebm
III	E
iv	Fm
bV	Gb
bv	Gbm
v	Gm
bVI	Ab
bvi	Abm
VI	A
bVII	Bb
bvii	Bbm
VII	B
vii	Bm

Minor Tonic


Cm Ebm Cm Cm F Cm Cm Abm Cm Cm Bbm Cm
i biii i i IV i i bvi i i bvii i
Cm Db Cm Cm Dbm Cm
i bII i i bii i
Cm D Cm Cm Dm Cm
i II i i ii i
Cm E Cm Cm Em Cm
i III i i iii i
Cm Gb Cm Cm Gbm Cm
i bV i i bv i
Cm A Cm Cm Am Cm
i VI i i vi i
Cm B Cm Cm Bm Cm
i VII i i vii i

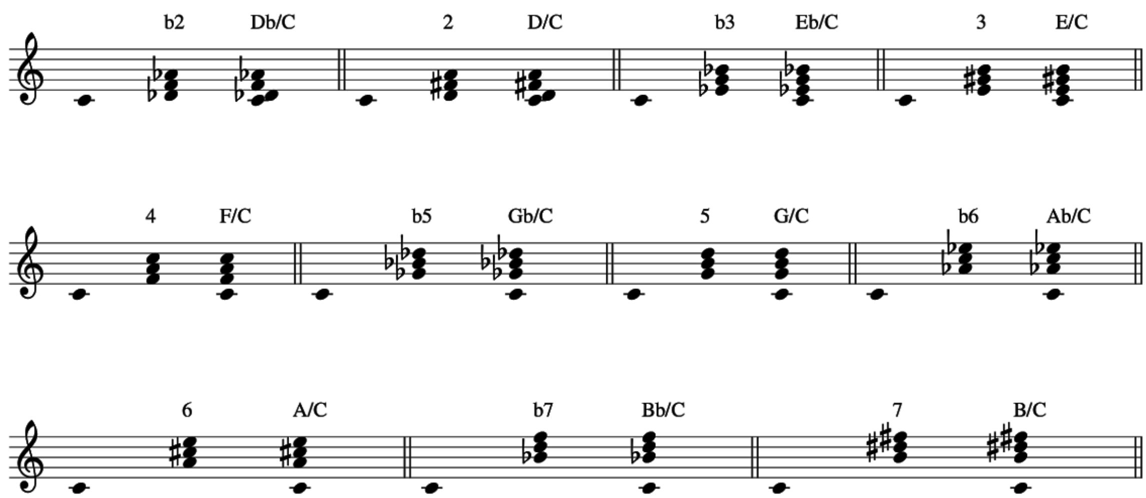
Modal Triads in C Minor

Roman Numeral	Triad
bII	Db
bii	Dbm
II	D
ii	Dm
biii	Ebm
III	E
iii	Em
IV	F
bV	Gb
bv	Gbm
bvi	Abm
VI	A
vi	Am
bvii	Bbm
VII	B
vii	Bm

Chapter 13: Bitonal Harmony

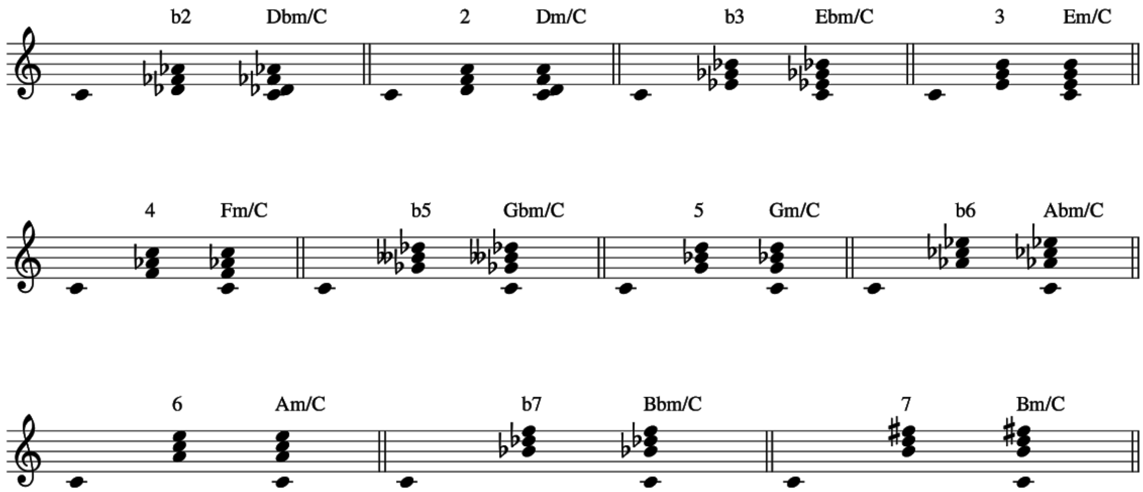
Bitonal Harmony includes voicings that pair a bass note with a Major or Minor triad built on another note. The resulting voicings are also known as “slash chords”, or “triads over bass notes”. In these examples, all triads are paired with a C bass note.

Major Triads



Root of Triad	Name	Harmonic Analysis
b2	Db/C	DbM7 (3rd inv) Phrygian
2	D/C	D7 (3rd inv) Lydian
b3	Eb/C	Cm7
3	E/C	CM7#5
4	F/C	F (2nd inv)
b5	Gb/C	C7b5 b9 Altered / Dom. Dim
5	G/C	CM9
b6	Ab/C	Ab (1st inv)
6	A/C	C13 b9 (Dom. Diminished)
b7	Bb/C	C9sus4
7	B/C	C°M7 (Tonic Dim)

Minor Triads



Root of Triad	Name	Harmonic Analysis
b2	Dbm/C	DbmM7 (3rd inv) C7b9 #5 (Altered)
2	Dm/C	Dm7 (3rd inv)
b3	Ebm/C	Cm7b5
3	Em/C	CM7
4	Fm/C	Fm (2nd inv)
b5	Gbm/C	C13 b9 b5 (Dom. Diminished)
5	Gm/C	C9
b6	Abm/C	Ab Min/Maj (2nd inv) Dominant Diminished
6	Am/C	Am (1st inv)
b7	Bbm/C	C7sus4 b9
7	Bm/C	CM9 #11